

The Splitting of the Idèle Class Group

Diego Rincón

Independent

Abstract

We prove that the idèle class group $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$ splits as a direct product $C_{\mathbb{Q}} \cong \mathbb{R}_+^* \times C_{\mathbb{Q}}^1$, where $C_{\mathbb{Q}}^1 = \ker(|\cdot|)$ is the norm-one subgroup. This splitting holds at the level of locally compact topological groups. It implies that $L^2(C_{\mathbb{Q}}, d^*x) \cong L^2(\mathbb{R}, dt) \otimes L^2(C_{\mathbb{Q}}^1)$, which proves Conjecture 3.1 of [1]: the operator $D = -i d/dt$ on $L^2(C_{\mathbb{Q}})$ is essentially self-adjoint with deficiency indices $(0,0)$. We identify precisely what additional hypothesis would then imply the Riemann Hypothesis.

1 Structure of the Idèle Class Group

Definition 1 (The idèle class group). The *ring of adèles* of $\mathbb{Q}$ is

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p,$$

the restricted direct product over all primes p . The *group of idèles* is $\mathbb{A}_{\mathbb{Q}}^* = \mathbb{R}^* \times \prod_p' \mathbb{Q}_p^*$. The *idèle class group* is $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$.

The *idèlic norm* is the continuous homomorphism

$$|\cdot| : C_{\mathbb{Q}} \rightarrow \mathbb{R}_+^*, \quad |x| = \prod_v |x_v|_v,$$

where v ranges over all places of $\mathbb{Q}$ (archimedean and non-archimedean). The product formula says $|\alpha| = 1$ for all $\alpha \in \mathbb{Q}^*$, so $|\cdot|$ is well-defined on $C_{\mathbb{Q}}$.

The *norm-one subgroup* is $C_{\mathbb{Q}}^1 = \ker(|\cdot| : C_{\mathbb{Q}} \rightarrow \mathbb{R}_+^*)$.

Theorem 2 (Compactness of $C_{\mathbb{Q}}^1$). $C_{\mathbb{Q}}^1$ is compact.

Proof. This is a standard result in algebraic number theory (see e.g. [3]). One argument: $C_{\mathbb{Q}}^1$ is the quotient of the norm-one idèles $\mathbb{A}_{\mathbb{Q}}^{*,1} = \{x \in \mathbb{A}_{\mathbb{Q}}^* : |x| = 1\}$ by $\mathbb{Q}^*$. By the product formula, $\mathbb{Q}^* \subset \mathbb{A}_{\mathbb{Q}}^{*,1}$. The quotient $\mathbb{A}_{\mathbb{Q}}^{*,1}/\mathbb{Q}^*$ is compact by the Hasse–Minkowski theorem and Minkowski’s theorem on lattice points (the number field analogue of the classical geometry-of-numbers argument). For $\mathbb{Q}$ specifically: $C_{\mathbb{Q}}^1 \cong \hat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$, the profinite completion of $\mathbb{Z}^*$, which is a compact profinite group. $\square$

2 The Splitting

Lemma 3 (Section of the norm map). *The norm map $|\cdot| : C_{\mathbb{Q}} \rightarrow \mathbb{R}_+^*$ splits: there exists a continuous group homomorphism $s : \mathbb{R}_+^* \rightarrow C_{\mathbb{Q}}$ such that $|\cdot| \circ s = \text{id}_{\mathbb{R}_+^*}$.*

Proof. Define $s(\lambda)$ to be the class of the idèle $(\lambda, 1, 1, 1, \dots) \in \mathbb{A}_{\mathbb{Q}}^*$: the idèle with $\lambda \in \mathbb{R}_+^*$ at the archimedean place and $1 \in \mathbb{Q}_p^*$ at every non-archimedean place p . Then $s(\lambda) \in C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$ is a well-defined element, and

$$|s(\lambda)| = |\lambda|_{\infty} \cdot \prod_p |1|_p = |\lambda| \cdot 1 = \lambda.$$

Continuity and the homomorphism property are immediate from the component-wise definitions. $\square$

Theorem 4 (Splitting of $C_{\mathbb{Q}}$). *There is an isomorphism of locally compact topological groups:*

$$C_{\mathbb{Q}} \cong \mathbb{R}_+^* \times C_{\mathbb{Q}}^1.$$

Explicitly: every $x \in C_{\mathbb{Q}}$ decomposes uniquely as $x = s(|x|) \cdot u$ with $u \in C_{\mathbb{Q}}^1$.

Proof. Define $\phi : C_{\mathbb{Q}} \rightarrow \mathbb{R}_+^* \times C_{\mathbb{Q}}^1$ by $\phi(x) = (|x|, x \cdot s(|x|)^{-1})$. The second component satisfies $|x \cdot s(|x|)^{-1}| = |x| \cdot |s(|x|)|^{-1} = |x|/|x| = 1$, so it lies in $C_{\mathbb{Q}}^1$. The map ϕ is a continuous group homomorphism. Its inverse is $(\lambda, u) \mapsto s(\lambda) \cdot u$, which is also continuous. Hence ϕ is an isomorphism of topological groups. $\square$

3 Self-Adjointness of D

Theorem 5 (Hilbert space decomposition). *Under the isomorphism $C_{\mathbb{Q}} \cong \mathbb{R}_+^* \times C_{\mathbb{Q}}^1$ and the coordinate $t = \log |\cdot| : \mathbb{R}_+^* \xrightarrow{\sim} \mathbb{R}$, the multiplicative Haar measure decomposes as $d^*x = dt \times d\mu$ where $d\mu$ is the (normalized) Haar measure on $C_{\mathbb{Q}}^1$. Therefore:*

$$L^2(C_{\mathbb{Q}}, d^*x) \cong L^2(\mathbb{R}, dt) \otimes L^2(C_{\mathbb{Q}}^1, d\mu).$$

Proof. The Haar measure on a direct product $G_1 \times G_2$ is the product of the Haar measures on each factor [4]. Under the splitting of Theorem 4, the Haar measure on $C_{\mathbb{Q}}$ is the product of the Haar measure $d^*\lambda = d\lambda/\lambda$ on $\mathbb{R}_+^*$ and the Haar measure $d\mu$ on $C_{\mathbb{Q}}^1$ (normalized to total mass 1). Under $t = \log \lambda$, $d^*\lambda = dt$. The L^2 isomorphism follows. $\square$

Corollary 6 (Essential self-adjointness of D on $C_{\mathbb{Q}}$). *The operator $D = -i d/dt \otimes 1$ on $L^2(C_{\mathbb{Q}}, d^*x) \cong L^2(\mathbb{R}, dt) \otimes L^2(C_{\mathbb{Q}}^1)$ is essentially self-adjoint with deficiency indices $(0, 0)$.*

Proof. By Theorem 5, $D = D_{\mathbb{R}} \otimes 1$ where $D_{\mathbb{R}} = -i d/dt$ on $L^2(\mathbb{R}, dt)$. The deficiency subspaces of D satisfy $\ker(D^* \mp i) = \ker(D_{\mathbb{R}}^* \mp i) \otimes L^2(C_{\mathbb{Q}}^1)$. By Theorem 3.1 of [1], $\ker(D_{\mathbb{R}}^* \mp i) = \{0\}$. Therefore $\ker(D^* \mp i) = \{0\}$, and D is essentially self-adjoint. $\square$

4 The Remaining Gap

Remark 7 (What self-adjointness gives and does not give). Corollary 6 proves that D is essentially self-adjoint on $L^2(C_{\mathbb{Q}}, d^*x)$. By the spectral theorem, D has real spectrum. The Fourier transform on $C_{\mathbb{Q}}$ decomposes $L^2(C_{\mathbb{Q}})$ into characters $\chi : C_{\mathbb{Q}} \rightarrow \mathbb{C}^*$, and D acts on the χ -isotypic component by $D|_{\chi} = -i \partial_t$ with spectral parameter $\lambda \in \mathbb{R}$.

The zeros of $\zeta(s)$ appear in the Weil explicit formula as the distribution:

$$W(h) = \hat{h}(0) + \hat{h}(1) - \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} (\hat{h}(k \log p) + \hat{h}(-k \log p)) + \cdots = \sum_{\rho} h(\rho).$$

Self-adjointness of D gives real continuous spectrum. The zeros of $\zeta(s)$ appear in this formula not as eigenvalues of D but as poles of the regularized resolvent of D after removing the contribution of $C_{\mathbb{Q}}^1$ (the compact part).

Theorem 8 (The precise remaining step). *The Riemann Hypothesis is equivalent to the following:*

For every non-negative smooth compactly supported function $h \geq 0$ on $\mathbb{R}$,

$$W(h) = \sum_{\rho} h\left(\frac{\rho-1/2}{i}\right) \geq 0,$$

where the sum is over non-trivial zeros ρ of $\zeta(s)$.

This positivity condition is not implied by self-adjointness of D alone. It requires an additional trace-positivity property of the action of D on the complement of the trivial spectrum. Self-adjointness is necessary but not sufficient. The gap is the positivity, not the self-adjointness.

Proof sketch. The implication $\text{RH} \Rightarrow W(h) \geq 0$ for $h \geq 0$: if all zeros have $\text{Re}(\rho) = 1/2$, then $\rho = 1/2 + i\gamma$ with $\gamma \in \mathbb{R}$, and $(\rho - 1/2)/i = \gamma \in \mathbb{R}$. For $h \geq 0$, the sum $\sum_{\rho} h(\gamma)$ is a sum of non-negative terms. The converse (positivity $\Rightarrow \text{RH}$) requires that $W(h) \geq 0$ for all $h \geq 0$ forces all $(\rho - 1/2)/i$ to be real, i.e., all zeros on the critical line. This follows from the uniqueness of the Fourier representation [2]. $\square$

Remark 9 (Summary of the mathematical situation). 1. D is essentially self-adjoint on $L^2(C_{\mathbb{Q}}, d^*x)$. **Proved above.**

2. The zeros of $\zeta(s)$ appear in the explicit formula as spectral data of D . **Connes [2].**

3. Self-adjointness gives real continuous spectrum of D . **Proved above.**

4. Real spectrum of D does *not* directly force real γ_n . **Noted above.**

5. $\text{RH} \Leftrightarrow$ positivity of $W(h)$ for $h \geq 0$. **Connes [2].**

6. The positivity $W(h) \geq 0$ for all $h \geq 0$. **Open.**

Item (6) is the Wall. It is a positivity statement. The Wall is not self-adjointness — that is settled. The Wall is positivity.

References

- [1] Rincón, D. (2026). The Deficiency Indices of D . Preprint.
- [2] Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica* 5(1), 29–106.
- [3] Cassels, J.W.S. and Fröhlich, A., eds. (1967). *Algebraic Number Theory*. Academic Press.
- [4] Folland, G.B. (1995). *A Course in Abstract Harmonic Analysis*. CRC Press.
- [5] Rincón, D. (2026). Week 10: Self-Adjointness. Preprint.
- [6] Rincón, D. (2026). The Arithmetic Wall. Preprint.